Project Proposal & Literature Review

Rate–Distortion Benchmarking Suite: Classical vs. Learned Image Codecs

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Abstract

This project develops a reproducible benchmarking pipeline comparing classical image codecs (JPEG, WebP, and AVIF) with pretrained learned image codecs through CompressAI. The comparison evaluates rate–distortion performance at matched or nearest-achievable bitrates using bits per pixel (bpp), PSNR, MS-SSIM, and LPIPS, while also recording encoding and decoding time. The evaluation uses the Kodak dataset and the CLIC validation set with consistent image preprocessing. A preliminary feasibility test has successfully executed a JPEG baseline and the pretrained `bmshj2018-hyperprior` model, providing an initial working foundation for the complete benchmark.

1 Problem Statement & Motivation

Image compression reduces the number of bits required to store or transmit an image while attempting to preserve its visual information and perceptual quality. Classical codecs such as JPEG, WebP, and AVIF use established compression algorithms, whereas learned image compression (LIC) models use neural networks to encode and reconstruct images. A fair comparison between these approaches is not straightforward: classical codecs are generally controlled by a quality factor (QF), while learned codecs use model-specific quality levels or rate–distortion operating points. Comparing settings with different labels does not guarantee that the codecs operate at similar bitrates. This project addresses the problem by developing a unified benchmarking suite that compares classical and learned codecs at matched or nearest-achievable bits per pixel (bpp), and evaluates reconstruction quality and computational cost. The project directly relates to Digital Image Processing topics including lossy image compression, image representation, image reconstruction, and objective image-quality assessment.

2 Related Work

Ballé *et al.* introduced a variational image-compression approach using a scale hyperprior, providing a foundation for learned codecs such as `bmshj2018-hyperprior` (1). CompressAI provides an open-source PyTorch implementation and evaluation platform for learned image-compression research, making it suitable for integrating pretrained models into a common experimental pipeline (2). Zhang *et al.* introduced LPIPS, a deep-feature-based perceptual metric that complements distortion measures such as PSNR (3).

Recent work reinforces the need to evaluate both compression quality and practical complexity. Minnen and Johnston studied the rate–distortion–computation trade-off in neural image compression and highlighted the importance of runtime and consistent evaluation conditions when comparing learned methods (4). Huang and Wu surveyed end-to-end learned image compression, discussing its progress relative to conventional codecs as well as the computational challenges that affect practical use (5). Chlubna and Zemcík presented a practical comparison of image compression methods across pixel formats and bit depths, using multiple quality metrics and encoding-time measurements (6). These studies support the proposed project's emphasis on bitrate-aware comparisons, multiple quality measures, and runtime reporting. The project will apply these principles in a reproducible pipeline focused on JPEG, WebP, AVIF, and pretrained CompressAI models.

Table 1: Key references and their relevance to the project.

Reference	**Year**	**Relevance to this project**
Ballé *et al.* (1)	2018	Introduces the scale-hyperprior approach used by the initial learned-codec baseline.
CompressAI (2)	2020	Provides the implementation framework and pretrained learned compression models.
Zhang *et al.* (3)	2018	Introduces LPIPS for perceptual image-quality evaluation.
Minnen and Johnston (4)	2023	Studies rate–distortion performance together with computational complexity.
Huang and Wu (5)	2024	Surveys end-to-end learned image compression and its practical challenges.
Chlubna and Zemcík (6)	2025	Compares practical image codecs using quality metrics and encoding time.

3 Dataset & Baseline Plan

Two datasets will be used. The Kodak dataset contains 24 lossless images, typically with dimensions of 768×512 or 512×768, and will provide a standard initial evaluation set. The second dataset is the CLIC validation set, which contains professional-quality images and provides broader image coverage. The project source specifies the validation split but does not record a separate numerical subset count; the benchmark will use the CLIC validation images obtained for the project. The dataset pages are the authoritative sources for access and applicable usage terms.

The classical baselines are JPEG, WebP, and AVIF. The learned baselines are pretrained CompressAI models, initially `bmshj2018-hyperprior` and `mbt2018-mean`. The initial feasibility test has already executed JPEG and `bmshj2018-hyperprior`; the remaining codecs and model will be integrated into the full benchmark.

Table 2: Dataset summary.

Dataset	**Subset used**	**Link**
Kodak	Complete set: 24 lossless images	`http://r0k.us/graphics/kodak/`
CLIC	Official validation set	`https://clic2025.compression.cc/tasks/`

Usage and licensing: The project will use the datasets according to the access and usage terms published by their respective providers. No separate license identifier was recorded in the initial project specification.

3.1 Bitrate-Matching Strategy

Classical codecs are controlled by quality factors, while learned codecs use model-specific quality levels. Therefore, the benchmark will match the resulting bitrate rather than equating quality settings. The protocol is:

1. Run each learned codec at predefined CompressAI quality levels, initially levels 1–6, and record the resulting bpp.
2. For each classical codec, search its quality-factor range for the setting whose bpp is closest to the learned codec's target bitrate.
3. Prefer matches within $\pm 1\%$ of the target bpp. If this tolerance cannot be achieved because of discrete quality settings, report the achieved bitrate and matching error.
4. Compare quality metrics at matched or nearest-achievable bitrate points, and explicitly document unmatched points.

4 Proposed Method

The proposed system is a modular Python benchmarking pipeline. Input images from Kodak and CLIC will be loaded and converted to a consistent RGB representation. Image dimensions will be padded or cropped according to the selected model's requirements. The selected preprocessing policy will be applied consistently across codecs to maintain a fair comparison.

The preprocessed images will be processed by two codec groups:

- **Classical codecs:** JPEG, WebP, and AVIF, using Python imaging libraries and the corresponding codec interfaces.
- **Learned codecs:** pretrained CompressAI models, initially `bmshj2018-hyperprior` and `mbt2018-mean`.

A bitrate-matching controller will search classical codec quality-factor settings to find the closest achievable bpp for each learned-codec operating point. The pipeline will calculate RGB PSNR, MS-SSIM, and LPIPS using the AlexNet backbone. Mean encoding and decoding time will be measured over 10 runs per image, excluding initial model loading. Hardware, software versions, and the warm-up policy will be recorded.

The project does not propose a new compression architecture. Its planned contribution is a controlled and reproducible comparison across classical and learned codecs, with explicit bitrate matching and multiple quality and runtime metrics. This approach is expected to provide a fairer comparison than evaluating codecs at arbitrary quality settings.

5 Feasibility & Compute Plan

A preliminary feasibility script, `benchmark_starter.py`, was executed successfully. It loaded a Kodak image, confirmed dimensions divisible by 64, and ran a classical JPEG baseline and the pretrained `bmshj2018-hyperprior` model. The recorded console output was:

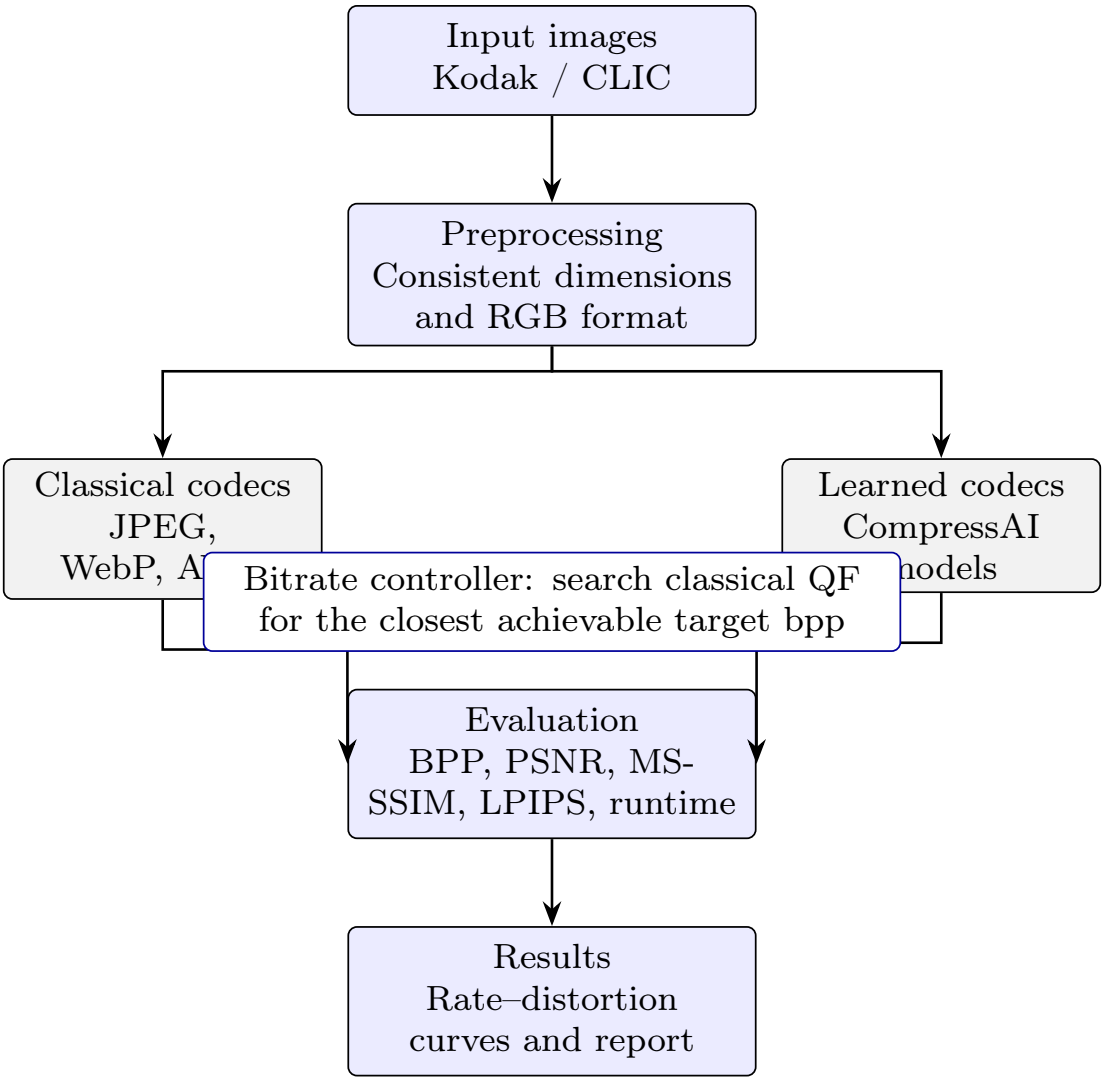


Figure 1: Benchmarking pipeline. The bitrate controller matches classical codec settings to learned-codec operating points as closely as the available quality settings permit.

```
================================================
Milestone 1: Feasibility Test & Pipeline Starter
================================================
[Preprocessing] Loaded image: 768x512
------------------------------------------------
[Classical JPEG] Quality: 75 | BPP: 1.8817
                 PSNR: 32.40 dB
[Learned Codec] bmshj2018-hyperprior (Q=3)
                 BPP: 0.5266 | PSNR: 28.68 dB
                 Encode time: 1.0882 s
================================================
[SUCCESS] Feasibility test passed.
================================================
```

These values confirm that both codec paths execute successfully. They are an initial feasibility check, not a bitrate-matched comparison, because the two codecs produced substantially different bpp values. The complete benchmark will use the matching procedure in Section 3.
Compute plan: The initial project specification does not identify a fixed CPU/GPU tier or record the hardware used for the feasibility run. Accordingly, the project will first establish a measured baseline on the available development machine, record its hardware and software configuration, and use the measured per-image runtime to estimate the full experiment duration. The observed JPEG/learned-codec pilot provides an initial timing reference; the recorded learned-codec encoding time was 1.0882 seconds for the tested image and setting. Full-dataset runtime and memory requirements will be measured during implementation rather than assumed in advance.

5.1 Minimum Viable Deliverable (MVP)

The minimum viable deliverable is a working benchmark on the 24 Kodak images that:

- Runs at least one classical codec (JPEG) and one pretrained learned codec (`bmshj2018-hyperprior`).
- Records bpp, RGB PSNR, and encoding time for both codec paths.
- Implements bitrate matching and reports the achieved bitrate difference, including cases outside the $\pm 1\%$ target.
- Produces a reproducible results table and rate–distortion plot.

The remaining codecs, additional metrics, and CLIC evaluation will extend this minimum working system.

5.2 Stretch Goal

After the MVP is working, the stretch goal is to run all planned classical codecs (JPEG, WebP, and AVIF), both initial CompressAI learned models, the CLIC validation set, and all planned metrics (PSNR, MS-SSIM, LPIPS, and encoding/decoding runtime). Additional learned codecs may be considered if the available project time and compute resources permit.

6 Risk Assessment

- **Codec or library compatibility issues:** Different codec interfaces or CompressAI dependencies may cause installation or inference failures. *Mitigation:* isolate codec adapters, keep the tested JPEG and `bmshj2018-hyperprior` paths as a fallback, and record working dependency versions.
- **Unmatched bitrate operating points:** Discrete quality settings may prevent a classical codec from reaching the target bpp within $\pm 1\%$. *Mitigation:* report the nearest achievable bitrate and matching error instead of discarding the result.
- **Dataset preprocessing differences:** Different padding or cropping policies could bias the comparison. *Mitigation:* define and document one consistent preprocessing policy for all codecs.
- **Experiment runtime:** Running multiple codecs, quality levels, metrics, and datasets may exceed the available project time. *Mitigation:* prioritize the Kodak-based MVP, then expand to CLIC and the remaining metrics.

7 Work Breakdown

The task allocation follows the team's stated responsibilities. Later milestone activities extend these responsibilities toward the complete benchmark.

Table 3: Milestone task allocation.

Milestone	Omar Sameh Mahmoud	David John Dawoud	Abdelrahman Refaat Saeed
M1 (Wk 3)	Dataset pipeline, classical codec baselines, and data integrity checks	CompressAI setup, model instantiation, and forward-pass compatibility	Bitrate-matching design, metric implementation, and timing protocol
M2 (Wk 6)	Complete dataset ingestion and classical codec integration	Integrate and test the planned learned codecs	Implement and validate bitrate matching and evaluation metrics
M3 (Wk 9)	Run final dataset and classical-codec experiments; verify outputs	Run learned-codec experiments and resolve model integration issues	Aggregate results, generate comparisons, and prepare final evaluation

Project repository: github.com/OmarSamehMahmoud/Rate-Distortion-Benchmarking-Suite

References

[1] J. Ballé, D. Minnen, S. Singh, S. Hwang, and N. Johnston, "Variational image compression with a scale hyperprior," in *International Conference on Learning Representations (ICLR)*, 2018.

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[3] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

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[6] T. Chlubna and P. Zemcík, "Comparative survey of image compression methods across different pixel formats and bit depths," *Signal, Image and Video Processing*, vol. 19, article 981, 2025, doi: 10.1007/s11760-025-04579-6.

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